

Part 0. Introduction

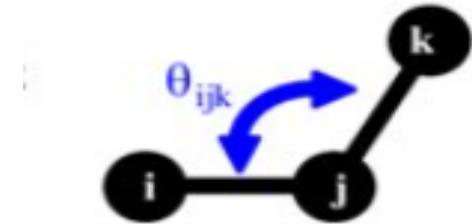
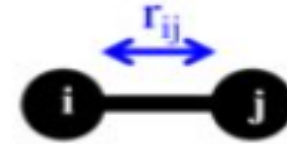
Plan

- (Semi-)empirical interatomic potentials: LJ, EAM, Tersoff, Coulomb, OPLS-AA, ReaxFF
- Advantages and disadvantages of (semi-)empirical interatomic potentials
- Machine-learning interatomic potentials: NNP, GAP, SNAP, MTP, ACE, DeepMD, PINN, NeQUIP, ETN
- Functional form of MTP
- Passive learning, training and validation errors, overfitting, uncertainty of estimation
- Comparison of different MLIPs
- MLIP-2 package
- Active learning
- Examples

Interatomic potentials

Interatomic potential is a mathematical function to calculate the potential energy of a system of N atoms. The general form is:

$$V = \sum_{i,j}^N V_2(r_{ij}) + \sum_{i,j,k}^N V_3(r_{ij}, r_{jk}, \theta_{ijk}) + \dots \quad (1)$$



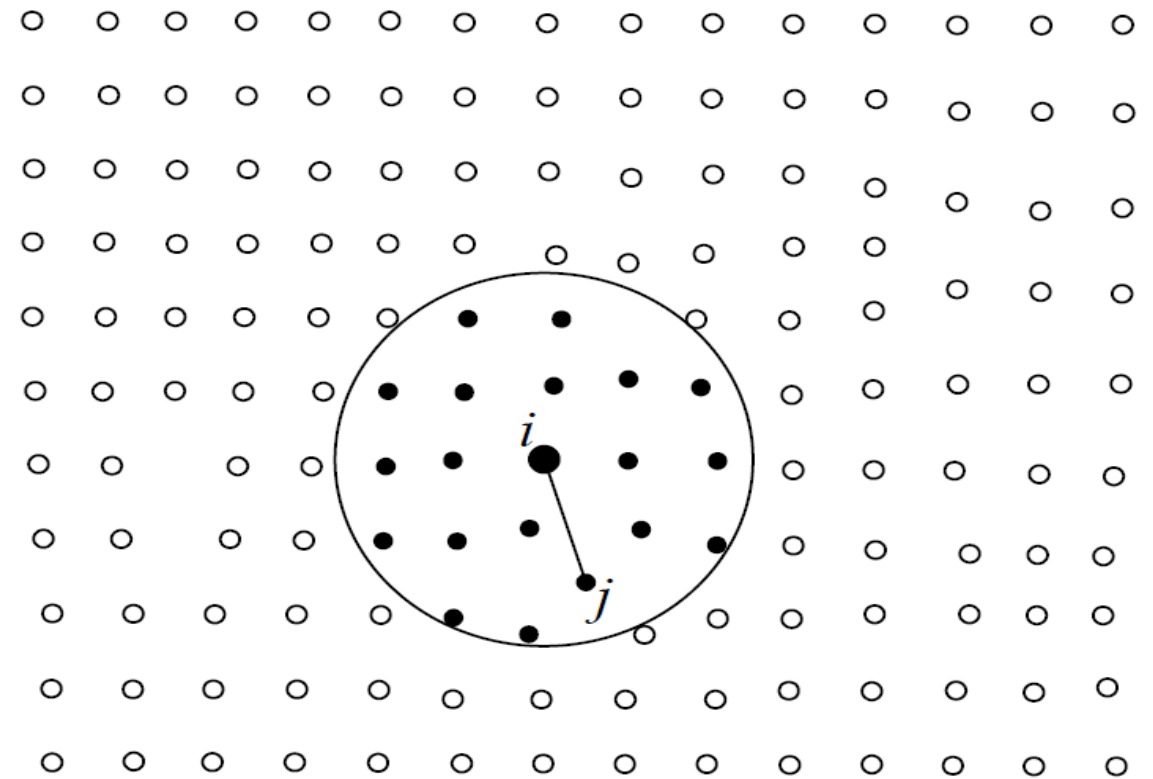
Here $r_{ij} = |\vec{r}_i - \vec{r}_j|$ is the distance between atoms and $\theta_{ijk} = \arccos \frac{(\vec{r}_{ij}, \vec{r}_{jk})}{r_{ij} \cdot r_{jk}}$ is the angle between the vectors $\vec{r}_{ij}$ and $\vec{r}_{jk}$.

V_2 term describes two-body interactions, V_3 term and the terms of the higher order describe many-body interactions.

Local interatomic potentials

We introduce the so-called cut-off radius R_{cut} and consider the atomic environments:

$$\mathbf{n}_i = \{r_{ij} : j = 1, \dots, N_{nb}^i\},$$



where the number of neighbors N_{nb}^i is determined by R_{cut} , i.e. $r_{ij} < R_{cut}$. The potential energy $V = \sum_{i=1}^n V_i = \sum_{i=1}^n V(\mathbf{n}_i)$. It should be smooth with respect to the atoms leaving and entering the environment and $V(r_{ij}) = 0$ if $r_{ij} \geq R_{cut}$.

(Semi-)empirical interatomic potentials

- Lennard-Jones potential (for noble gases and van der Waals interactions)
- Embedded atom model (for metals and alloys)
- Tersoff potential (for semiconductors and insulators)
- Coulomb potential (for long-range interactions)
- OPLS-AA (for organic molecules; is not applicable to chemical reactions)
- ReaxFF (for molecular systems; applicable to chemical reactions)

Advantages and disadvantages of (semi-)empirical potentials

Advantages:

- fast (can be applied for the systems of billion atoms)
- have a physical meaning

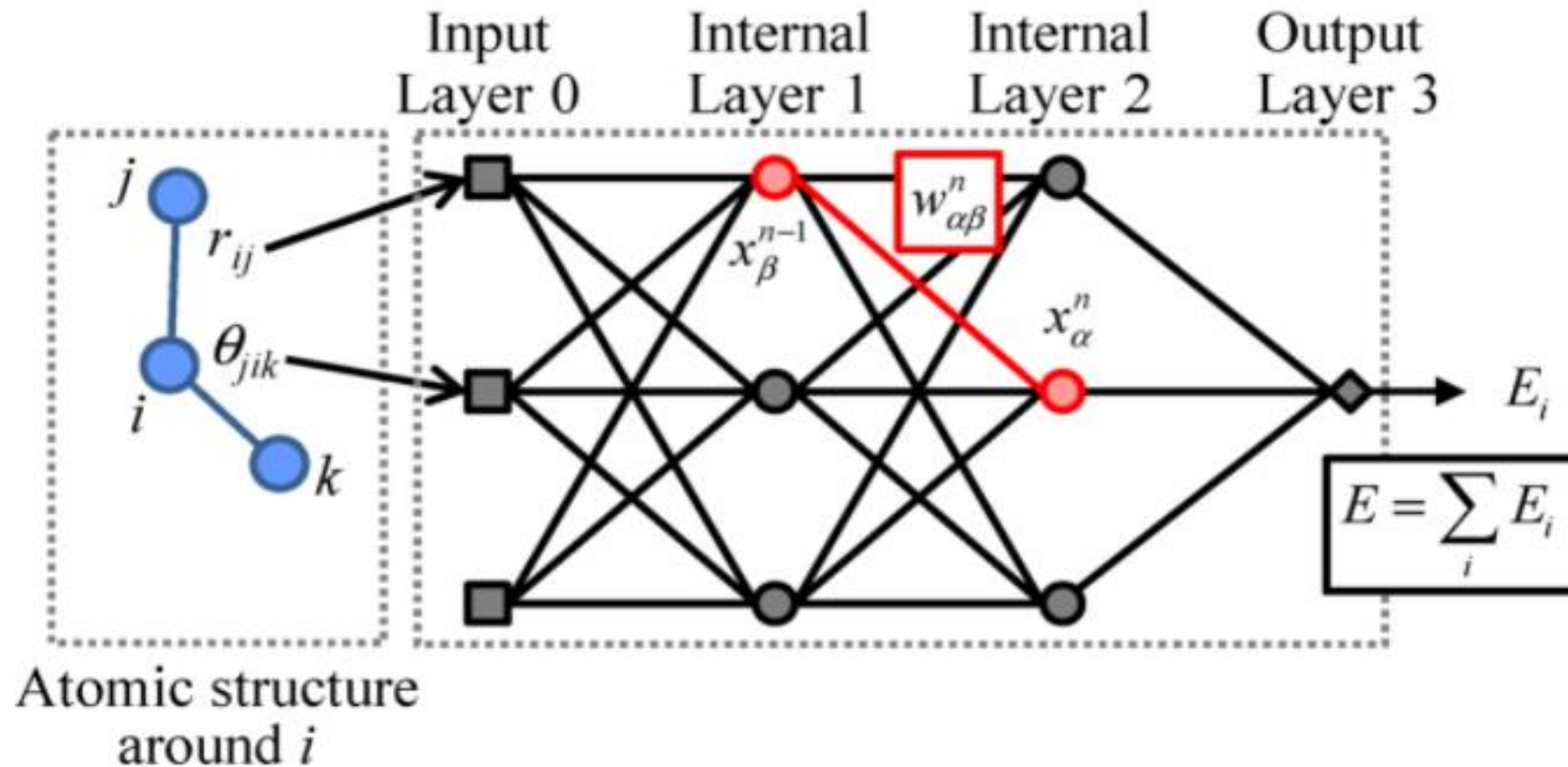
Disadvantages:

- not accurate enough (typically parametrized to DFT data, but are not able to predict properties of materials with DFT accuracy)
- applicable only to specific materials under specific conditions (problems with transferability)
- mainly used only for materials with small number of components

Alternative: machine-learning interatomic potentials!

Machine-learning interatomic potentials (MLIPs): general idea

Atomic positions are fed into the **input layer**, the **output layer** delivers the **energy**, and the **hidden layers** inserted in between provide **additional adjustable parameters** and enhance the **flexibility** of the model.



$$\sum_{k=1}^K \left[w_e \left(E_k^{\text{DFT}} - E_k^{\text{MLIP}}(\boldsymbol{\theta}) \right)^2 + w_f \sum_{i=1}^n \left| f_{i,k}^{\text{DFT}} - f_{i,k}^{\text{MLIP}}(\boldsymbol{\theta}) \right|^2 \right] \rightarrow \min$$

Machine-learning interatomic potentials

- Neural Network Potential (**NNP**) [Behler, J., & Parrinello, M. (2007). Generalized neural-network representation of high-dimensional potential-energy surfaces. *Physical review letters*, 98(14), 146401]
- Gaussian approximation potential (**GAP**) [Bartók, A. P., Payne, M. C., Kondor, R., & Csányi, G. (2010). Gaussian approximation potentials: The accuracy of quantum mechanics, without the electrons. *Physical review letters*, 104(13), 136403]
- Spectral neighbor analysis potential (**SNAP**) [Thompson, A. P., Swiler, L. P., Trott, C. R., Foiles, S. M., & Tucker, G. J. (2015). Spectral neighbor analysis method for automated generation of quantum-accurate interatomic potentials. *Journal of Computational Physics*, 285, 316-330]
- **Moment Tensor Potential (MTP)** [Shapeev, A. V. (2016). Moment tensor potentials: A class of systematically improvable interatomic potentials. *Multiscale Modeling & Simulation*, 14(3), 1153-1173]
- **DeepMD** [Wang, H., Zhang, L., Han, J., & Weinan, E. (2018). DeePMD-kit: A deep learning package for many-body potential energy representation and molecular dynamics. *Computer Physics Communications*, 228, 178-184]
- Atomic Cluster Expansion (**ACE**) [Drautz, R. (2019). Atomic cluster expansion for accurate and transferable interatomic potentials. *Physical Review B*, 99(1), 014104]
- Physically informed neural network (**PI-NN**) [Pun, G. P., Batra, R., Ramprasad, R., & Mishin, Y. (2019). Physically informed artificial neural networks for atomistic modeling of materials. *Nature communications*, 10(1), 2339]
- Neural Equivariant Interatomic Potentials (**NequIP**) [Batzner, S., Musaelian, A., Sun, L., Geiger, M., Mailoa, J. P., Kornbluth, M., ... & Kozinsky, B. (2022). E (3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials. *Nature communications*, 13(1), 2453]
- Equivariant Tensor Network (**ETN**) [Hodapp, M., & Shapeev, A. (2024). Equivariant tensor network potentials. *Machine Learning: Science and Technology*, 5(3), 035075]
- ...

Part 1. Moment Tensor Potential

Moment Tensor Potential

MTP was proposed in [Multiscale Model. Simul., Vol. 14, No. 3, pp. 1153-1173]. This potential is **local**, i.e. its energy E^{MTP} is the sum of contributions $V(\mathbf{n}_i)$ of individual atomic neighborhoods $\mathbf{n}_i$, $i = \overline{1, n}$:

$$E^{MTP} = \sum_{i=1}^n V(\mathbf{n}_i).$$

The function V is **invariant** to **atomic permutations**, **rotations**, and **reflections**, it is **smooth with respect to atoms leaving and entering the interaction neighborhood**. This function is linearly expanded through a set of **basis functions** B_α :

$$V(\mathbf{n}_i) = \sum_{\alpha} \xi_{\alpha} B_{\alpha}(\mathbf{n}_i),$$

where $\xi = \{\xi_{\alpha}\}$ is the set of **linear parameters**.

Moment Tensor Descriptor (1)

Denote $\mathbf{n}_i = (\{r_{i1}, z_i, z_1\}, \dots, \{r_{ij}, z_i, z_j\}, \dots, \{r_{iN_{Nb}}, z_i, z_{N_{Nb}}\})$, where r_{ij} are relative atomic positions, z_i, z_j are the types of central and neighboring atoms, Nb is the number of neighbors. **Moment Tensor Descriptor:**

$$M_{\mu, \nu}(\mathbf{n}_i) = \sum_j f_{\mu}(|r_{ij}|, z_i, z_j) \underbrace{r_{ij} \otimes \dots \otimes r_{ij}}_{\nu \text{ times}},$$

which consists of the **radial part**:

$$f_{\mu}(|r_{ij}|, z_i, z_j) = \sum_{\beta} c_{\mu, z_i, z_j}^{(\beta)} T_{\beta}(|r_{ij}|) (R_{cut} - |r_{ij}|)^2,$$

where $\mathbf{c} = \{c_{\mu, z_i, z_j}^{(\beta)}\}$ is the set of **radial parameters**, T_{β} are Chebyshev polynomials, and the **angular part** $r_{ij} \otimes \dots \otimes r_{ij}$, where “ $\otimes$ ” is the outer product, e.g., if $\nu = 2$:

$$r_{ij} \otimes r_{ij} = \begin{pmatrix} x_{ij}^2 & x_{ij}y_{ij} & x_{ij}z_{ij} \\ y_{ij}x_{ij} & y_{ij}^2 & y_{ij}z_{ij} \\ z_{ij}x_{ij} & z_{ij}y_{ij} & z_{ij}^2 \end{pmatrix}.$$

Moment Tensor Descriptor (2)

To illustrate that $r_{ij} \otimes \cdots \otimes r_{ij}$ is an angular part, assume that the vectors r_{ij} are two-dimensional and that we can express them in polar coordinate (ρ, θ) centered at the i -th atom.

Tensor of the first order: $\mathbf{r}_{ij}^{\otimes 1} = \mathbf{r}_{ij} = |\mathbf{r}_{ij}|(\cos \theta_{ij}, \sin \theta_{ij})$

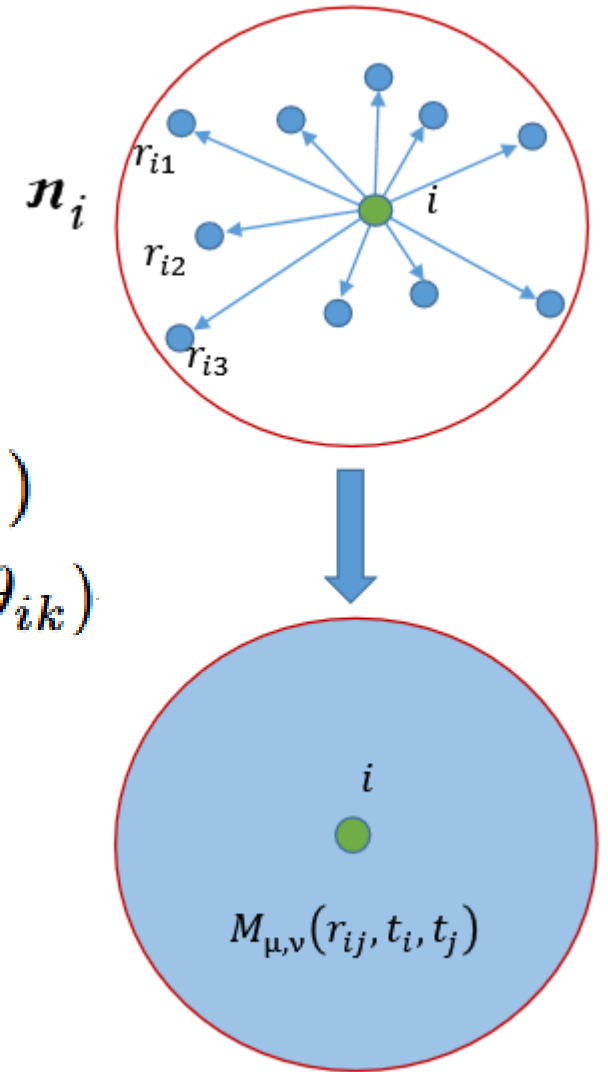
Contraction yielding a scalar: $\mathbf{r}_{ij} \cdot \mathbf{r}_{ik} = |\mathbf{r}_{ij}| |\mathbf{r}_{ik}| \cos(\theta_{ij} - \theta_{ik})$

Tensor of the second order:

$$\mathbf{r}_{ij}^{\otimes 2} = \mathbf{r}_{ij} \mathbf{r}_{ij}^{\top} = |\mathbf{r}_{ij}|^2 \begin{pmatrix} \cos^2 \theta_{ij} & \sin \theta_{ij} \cos \theta_{ij} \\ \sin \theta_{ij} \cos \theta_{ij} & \sin^2 \theta_{ij} \end{pmatrix}$$

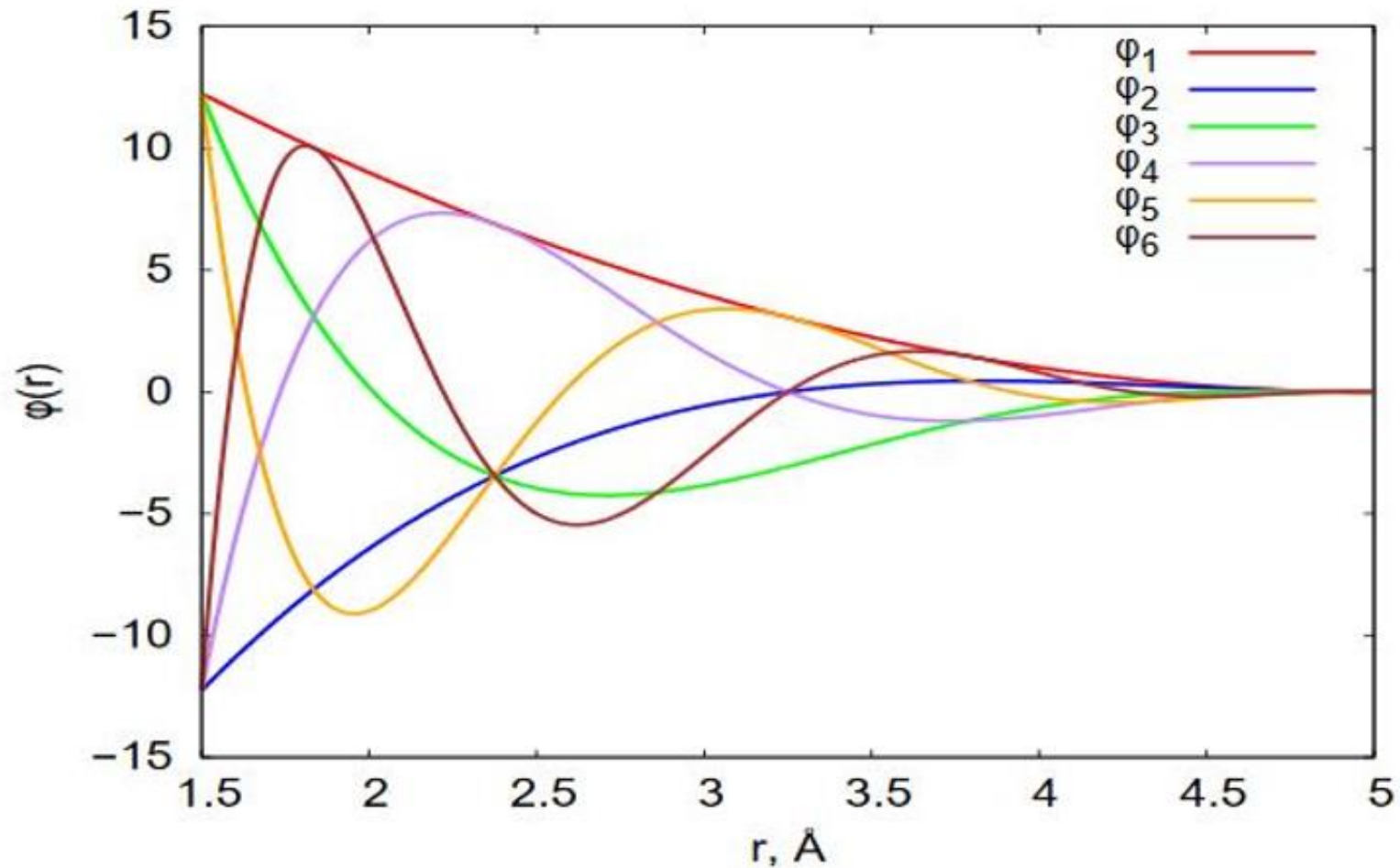
Contraction:

$$\mathbf{r}_{ij}^{\otimes 2} : \mathbf{r}_{ik}^{\otimes 2} = |\mathbf{r}_{ij}|^2 |\mathbf{r}_{ik}|^2 \cos^2(\theta_{ij} - \theta_{ik})$$



Radial basis example

$$\varphi_{\beta}(r) = \begin{cases} T_{\beta}(r)(R_{cut} - r)^2, & r < R_{cut} = 5 \text{ \AA} \\ 0, & r \geq R_{cut} = 5 \text{ \AA} \end{cases}$$



Level of Moment Tensor Descriptor

Level of Moment Tensor Descriptor:

$$\text{lev } M_{\mu,\nu} = 2 + 4\mu + \nu$$

Level of Moment Tensor Descriptor Product:

$$\text{lev } \prod_{p=1}^P M_{\mu_p,\nu_p} = \sum_{p=1}^P (2 + 4\mu_p + \nu_p)$$

Examples:

$$\text{lev } M_{0,0} = 2, \text{lev } M_{0,1} = 3, \text{lev } M_{1,1} = 7, \text{lev } M_{0,2} = 4$$

$$\text{lev } M_{1,0}^2 = 12, \text{lev } M_{0,0}^4 = 8, \text{lev } M_{2,0}^3 = 30$$

$$\text{lev } (M_{1,1} \cdot M_{0,1}) = 10, \text{lev } (M_{1,2} : M_{0,2}) = 12, \text{lev } ((M_{0,3} M_{0,2}) \cdot M_{0,1}) = 12$$

Basis functions

Basis function B_α is a contraction of any number of Moment Tensor Descriptors, yielding a scalar, e.g.:

$$M_{\mu,0}; M_{\mu,1} \cdot M_{\mu,1}; M_{\mu,2} : M_{\mu,2},; (M_{\mu,2} \cdot M_{\mu,1}) \cdot M_{\mu,1},; \dots$$

Thus, level of basis function B_α :

$$\text{lev} B_\alpha = \text{lev} \prod_{p=1}^P M_{\mu_p, \nu_p} = \sum_{p=1}^P (2 + 4\mu_p + \nu_p) .$$

To define an MTP we choose some $\text{lev}_{\max}$ and include in basis any function B_α with $\text{lev} B_\alpha \leq \text{lev}_{\max}$.

Example: MTP of level 8

Let $\text{lev}_{\max} = 8$. Then we have 9 basis functions B_α :

$$B_1 = M_{0,0}; \text{lev}M_{0,0} = 2 \leq \text{lev}_{\max}$$

$$B_2 = M_{1,0}; \text{lev}M_{1,0} = 6 \leq \text{lev}_{\max}$$

$$B_3 = M_{0,0}^2; \text{lev}M_{0,0}^2 = 4 \leq \text{lev}_{\max}$$

$$B_4 = M_{0,1} \cdot M_{0,1}; \text{lev}(M_{0,1} \cdot M_{0,1}) = 6 \leq \text{lev}_{\max}$$

$$B_5 = M_{0,2} : M_{0,2}; \text{lev}(M_{0,2} : M_{0,2}) = 8 \leq \text{lev}_{\max}$$

$$B_6 = M_{0,0}M_{1,0}; \text{lev}(M_{0,0}M_{1,0}) = 8 \leq \text{lev}_{\max}$$

$$B_7 = M_{0,0}^3; \text{lev}M_{0,0}^3 = 6 \leq \text{lev}_{\max}$$

$$B_8 = M_{0,0}(M_{0,1} \cdot M_{0,1}); \text{lev}(M_{0,0}(M_{0,1} \cdot M_{0,1})) = 8 \leq \text{lev}_{\max}$$

$$B_9 = M_{0,0}^4; \text{lev}M_{0,0}^4 = 8 \leq \text{lev}_{\max}$$

Passive training (fitting) of MTP

Let K be a number of configurations in a **training set** with DFT energies, forces, and stresses. Denote a set of **MTP parameters** to be found by $\boldsymbol{\theta}$. The **fitting** consists of finding the parameters $\boldsymbol{\theta}$ that minimize the following **loss function**:

$$\sum_{k=1}^K \left[w_e \left(E_k^{\text{DFT}} - E_k^{\text{MTP}}(\boldsymbol{\theta}) \right)^2 + w_f \sum_{i=1}^n |f_{i,k}^{\text{DFT}} - f_{i,k}^{\text{MTP}}(\boldsymbol{\theta})|^2 + w_s \sum_{i=1}^6 (\sigma_{i,k}^{\text{DFT}} - \sigma_{i,k}^{\text{MTP}}(\boldsymbol{\theta}))^2 \right] \rightarrow \min$$

where w_e , w_f , and w_s are non-negative weights.

In practice: $w_e = 1$, $w_f = 0.01$, $w_s = 0.001$

[Novoselov, I. I., et al. "Moment tensor potentials as a promising tool to study diffusion processes." *Computational Materials Science* 164 (2019): 46-56]

Training errors

After fitting of MLIP we compute **training errors**:

$$\text{RMSE}_{\text{tr}}(E)^2 = \frac{1}{K} \sum_{k=1}^K \left(\frac{E_k^{\text{DFT}}}{n} - \frac{E_k^{\text{MTP}}(\boldsymbol{\theta})}{n} \right)^2 ,$$

$$\text{RMSE}_{\text{tr}}(f)^2 = \frac{1}{K} \sum_{k=1}^K \frac{1}{3n} \sum_{i=1}^n |f_{i,k}^{\text{DFT}} - f_{i,k}^{\text{MTP}}(\boldsymbol{\theta})|^2 ,$$

$$\text{RMSE}_{\text{tr}}(\sigma)^2 = \frac{1}{K} \sum_{k=1}^K \frac{1}{6} (\sigma_{i,k}^{\text{DFT}} - \sigma_{i,k}^{\text{MTP}}(\boldsymbol{\theta}))^2 .$$

Validation of MLIP and overfitting

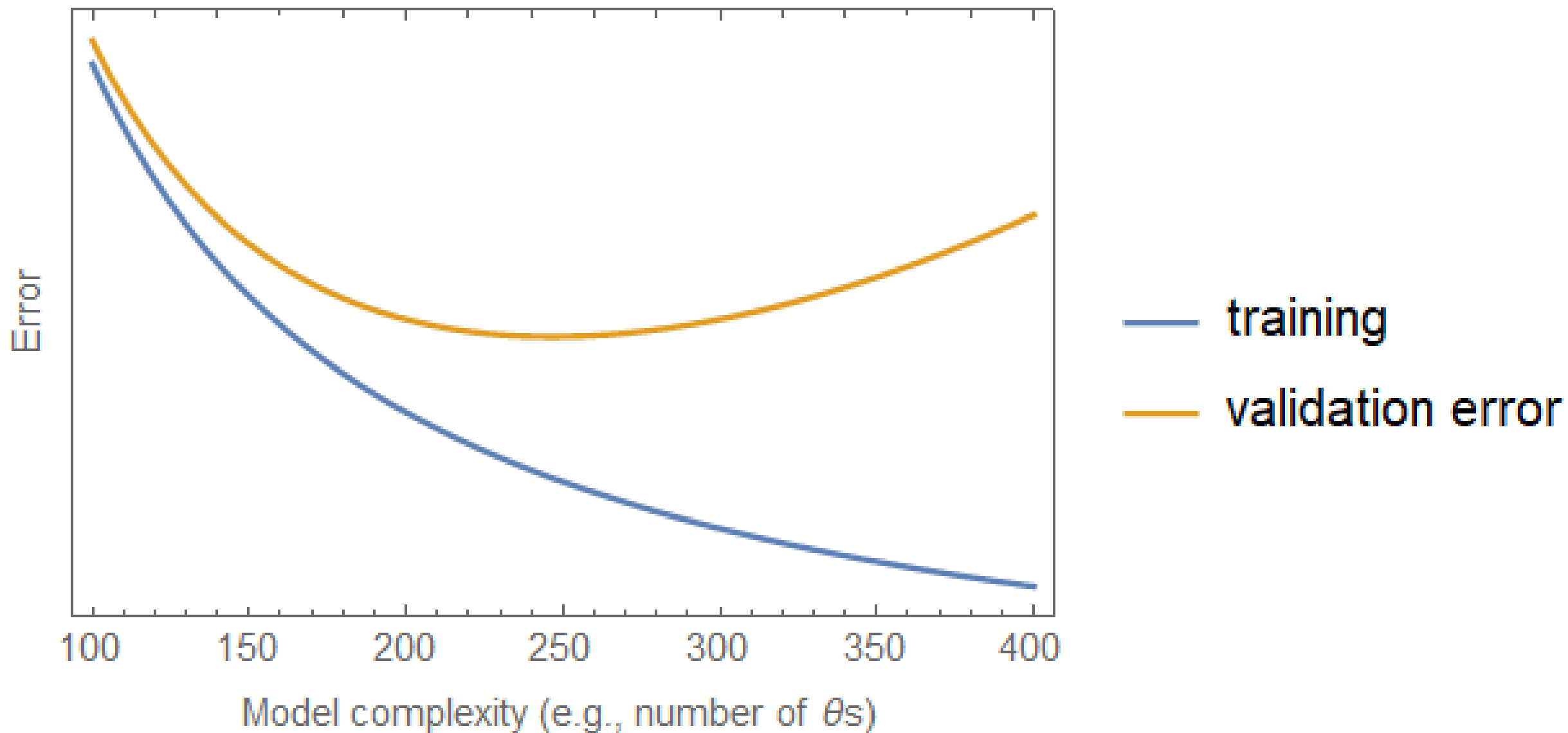
In order to estimate the **quality** of the trained MLIP we compute **validation errors**. For this aim we construct a **validation set** of M configurations that are not included in the **training set** and are not correlated with the ones in the **training set**. Typically $M/K \approx 0.2$. Next we calculate the **validation errors**.

Overfitting: $\text{RMSE}_{\text{tr}}(E)^2 \ll \text{RMSE}_{\text{vld}}(E)^2$ (e.g., the **number of parameters in MLIP** is **greater** than the **number of configurations in the training set**)

Underfitting: both $\text{RMSE}_{\text{tr}}(E)^2$ and $\text{RMSE}_{\text{vld}}(E)^2$ are **too big** (e.g., not enough parameters in MLIP)

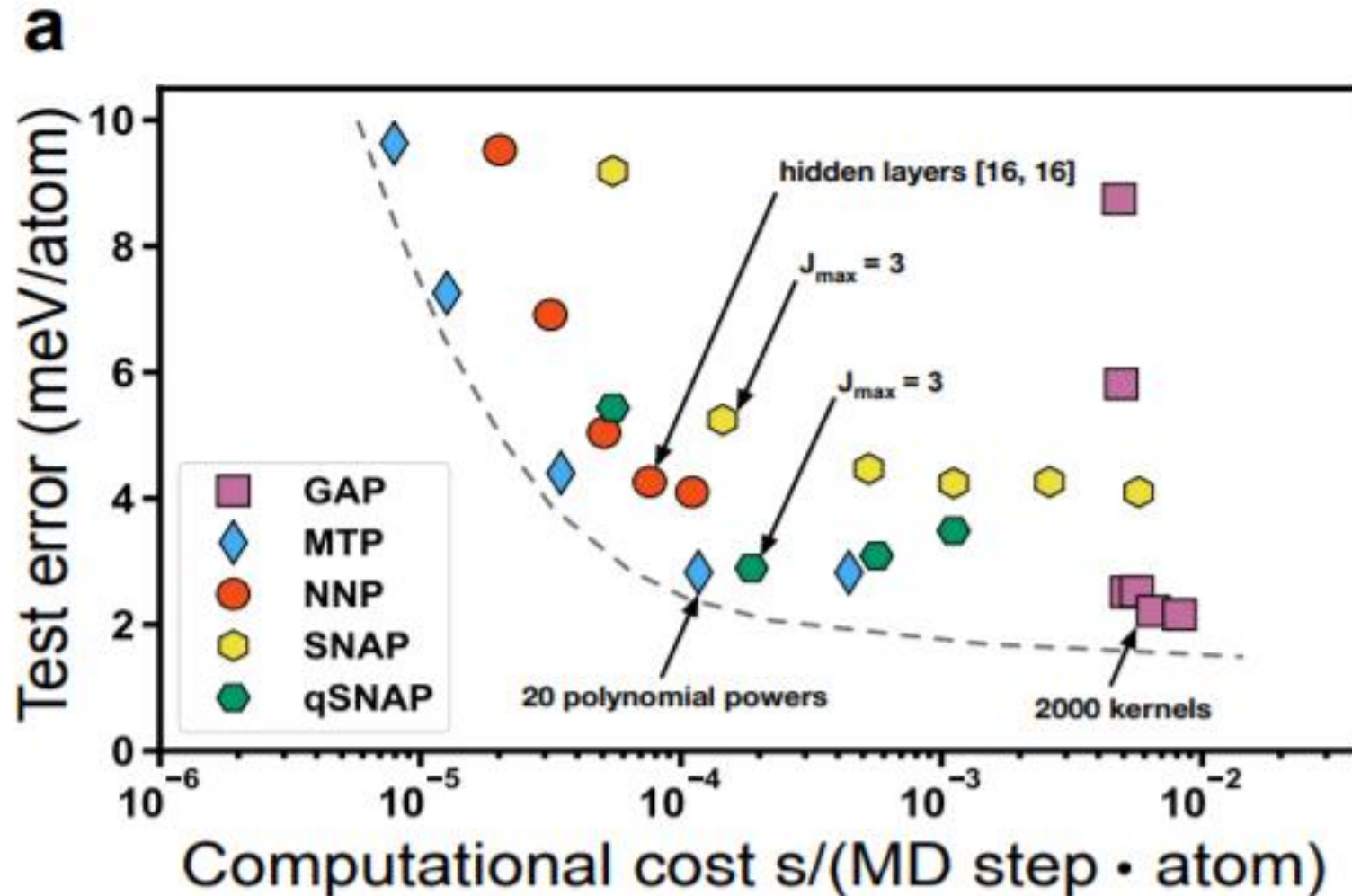
Good fitting: $\text{RMSE}_{\text{tr}}(E)^2 \approx \text{RMSE}_{\text{vld}}(E)^2$ and **these errors are small** enough

Validation of MLIP: illustration



Comparison of different MLIPs: performance

In [J. Phys. Chem. A 2020, 124, 4, 731–745] NNP, GAP, SNAP, qSNAP, and MTP were trained on the same datasets and compared to each other in terms of [performance](#) and [accuracy](#).



MLIP-2 package

Developers: Alexander V. Shapeev, Evgeny V. Podryabinkin, Konstantin Gubaev, and Ivan S. Novikov

Moment Tensor Potentials and algorithms for their training are implemented in the MLIP-2 package

MLIP-2 is an open-source code available at:

<https://gitlab.com/ashapeev/mlip-2>

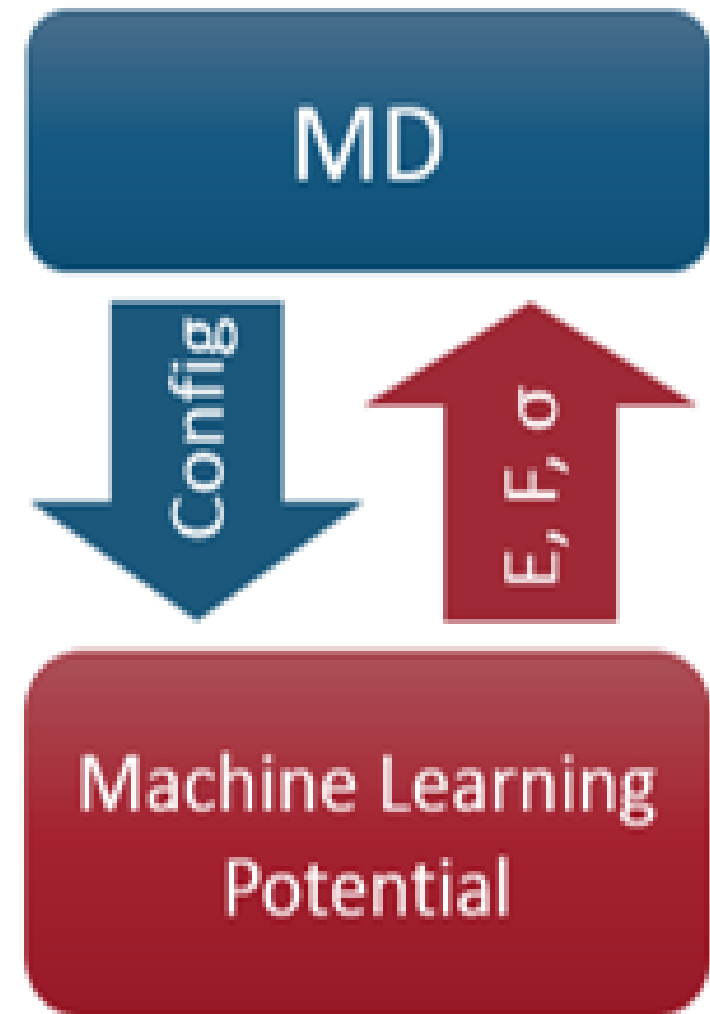
An interface between LAMMPS and MLIP-2 available at:

<https://gitlab.com/ashapeev/interface-lammps-mlip-2>

Part 2. Active Learning

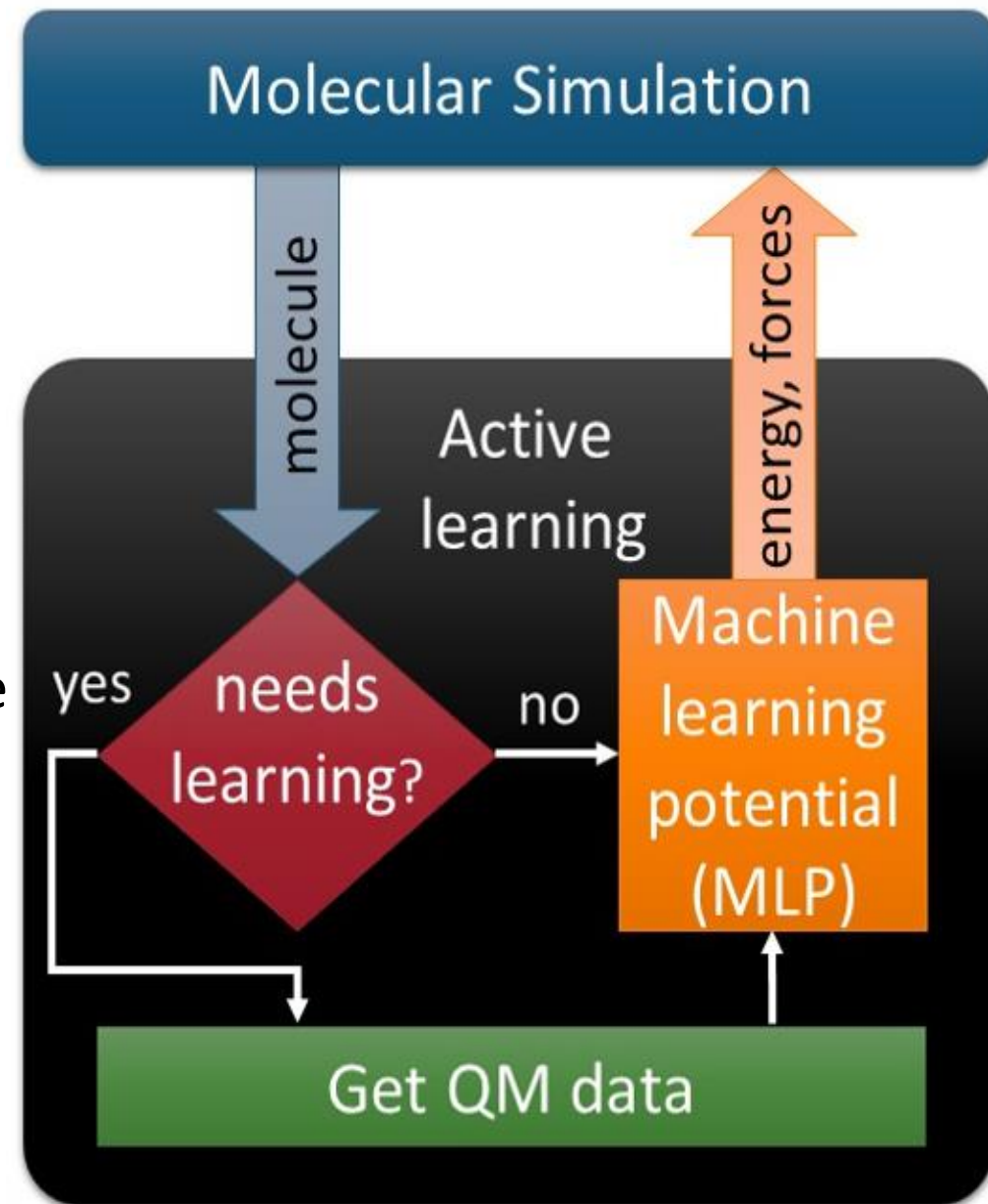
Passive learning

- Training set is being created **off-line** (e.g., random perturbations of atoms, non-correlated configurations from MD, etc.)
- Training set can cover only a **small part of the configuration space**
- MLIP trained on such the training set can have **problems with transferability** and often has “artifacts”
- Training set can contain “unnecessary” configurations and, thus, **“unnecessary” ab initio calculations** could be conducted
- Simulation (e.g., MD) runs with **the same MLIP**



Active learning

- Training set is being created **on-line** (i.e., during any simulation like MD, relaxation, etc.)
- **MLIP** “participates” in **constructing** the training set
- Training set **covers almost the whole configuration space**
- **MLIP** created during the active learning is **transferable** and does not have “artifacts”
- **Active learning** allows **reducing a number of** expensive **ab initio** calculations
- **MLIP** is being **updated** after each active learning step

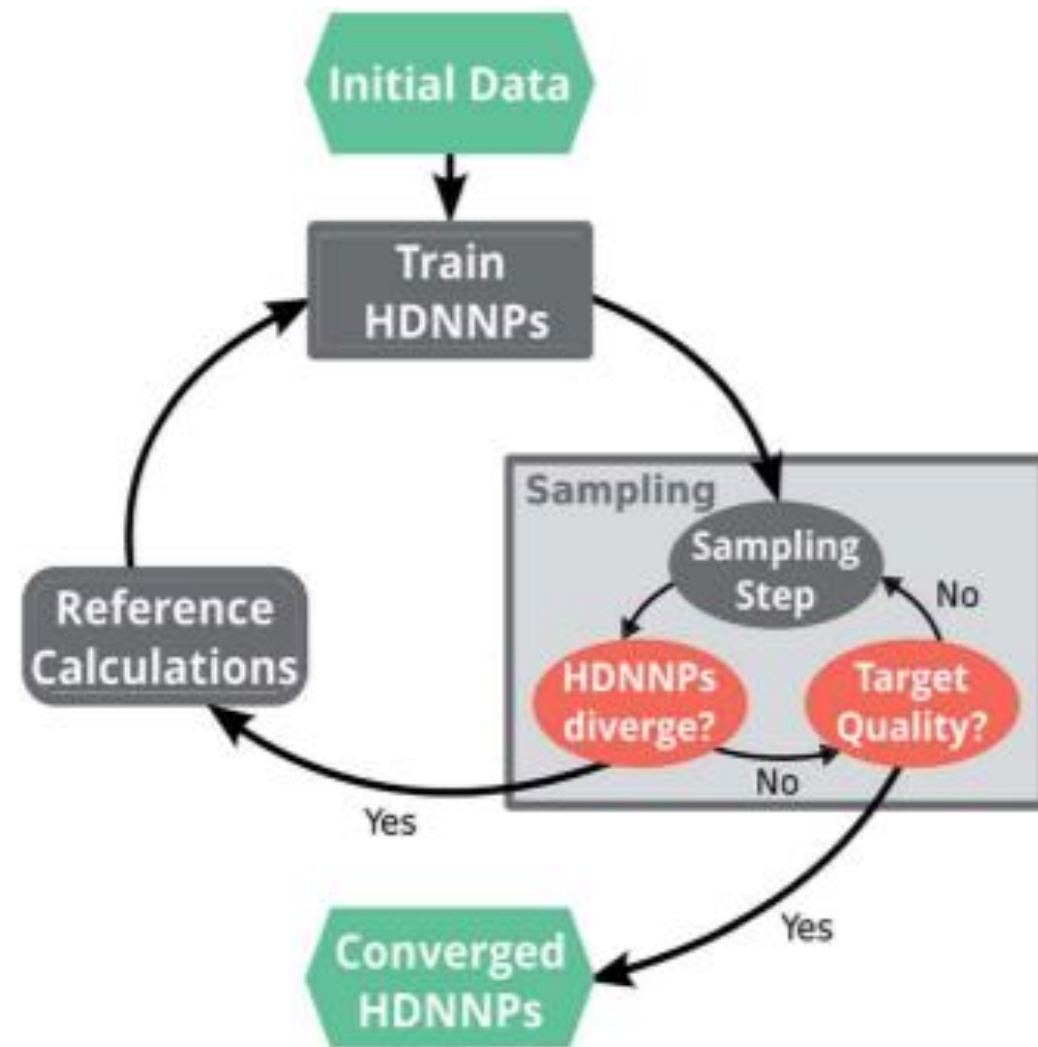


Example: active learning with an ensemble of MLIPs

In the paper [Chem. Sci., 2017, 8, 6924] the active learning algorithm with the ensemble of NNPs was proposed.

Algorithm

1. Create **initial training set** and train an **ensemble** of NNPs.
2. Run **sampling step**: calculate energy and forces for the current configuration. If the **prediction uncertainty (standard deviation)** of NNPs ensemble is **too high**, then we **stop sampling step**.
3. Run **reference (ab initio) calculations** for the **selected configurations**, **update training set**, **re-train the ensemble** of NNPs.
4. Run steps 2-3 **until no problem configuration** is found (i.e., the standard deviation is small enough).

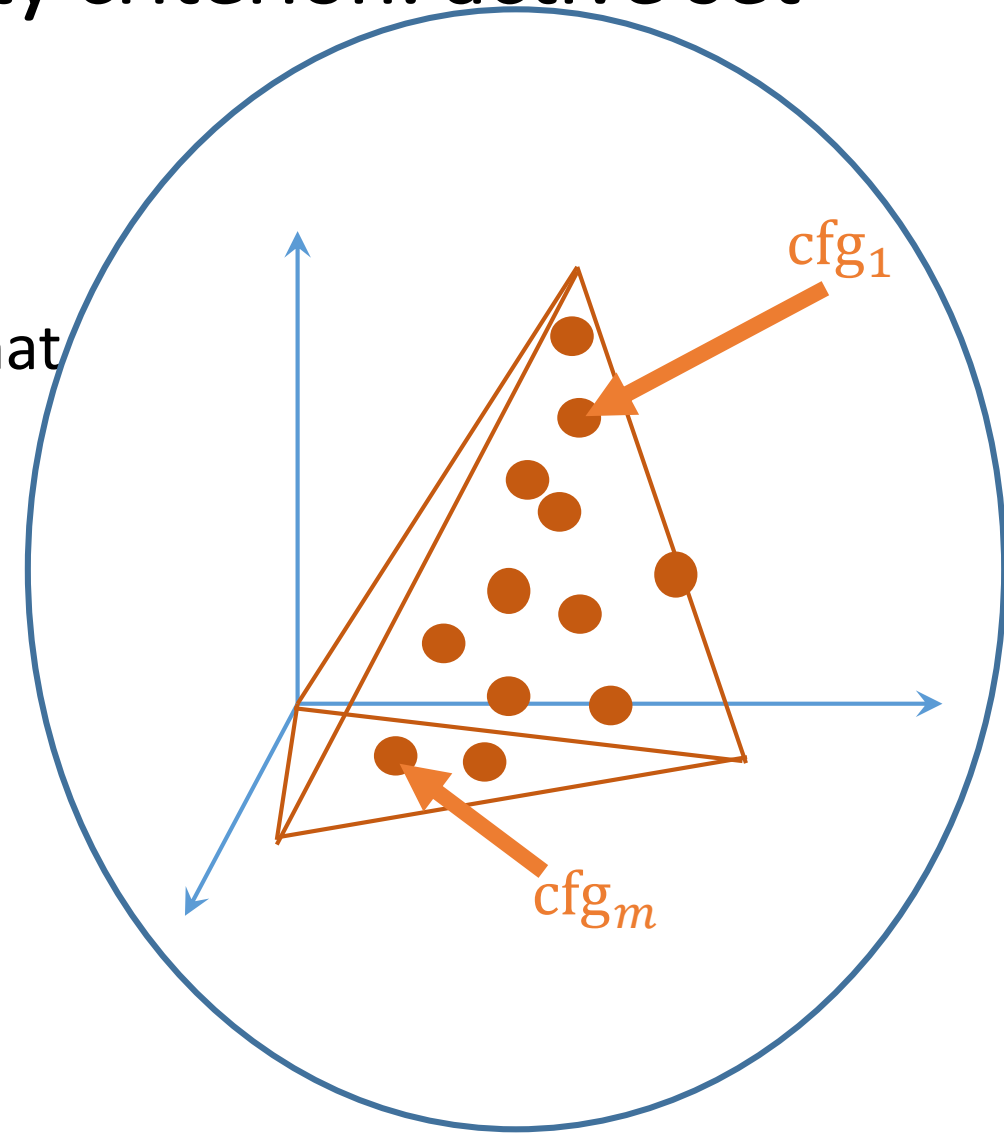


Active learning (AL) with D-optimality criterion: active set

Assume we have a **training set** of K configurations ($\text{cfg}_1, \dots, \text{cfg}_K$) and the potential with m **optimized parameters** $\boldsymbol{\theta} = (\theta_1, \dots, \theta_m)$. Denote the potential energy by $E = E(\boldsymbol{\theta}, \text{cfg})$. By **active set** we define a **subset** of size m configurations from the **training set** that **maximizes the determinant (volume)** of the matrix:

$$A = \begin{bmatrix} \frac{\partial E}{\partial \theta_1}(\boldsymbol{\theta}, \text{cfg}_1) & \dots & \frac{\partial E}{\partial \theta_m}(\boldsymbol{\theta}, \text{cfg}_1) \\ \vdots & \ddots & \vdots \\ \frac{\partial E}{\partial \theta_1}(\boldsymbol{\theta}, \text{cfg}_m) & \dots & \frac{\partial E}{\partial \theta_m}(\boldsymbol{\theta}, \text{cfg}_m) \end{bmatrix}$$

For the **case** $K > m$ in order to find the **active set** we use the so-called **MaxVol algorithm**. If $K \leq m$ we **include some of (or, even, all) K configurations** in the **active set** and fill in the rest of the lines with **zeros outside the diagonal** and with **ones in the diagonal**.



Pyramid illustrates an **active set**.
Blue circle illustrates configuration space.

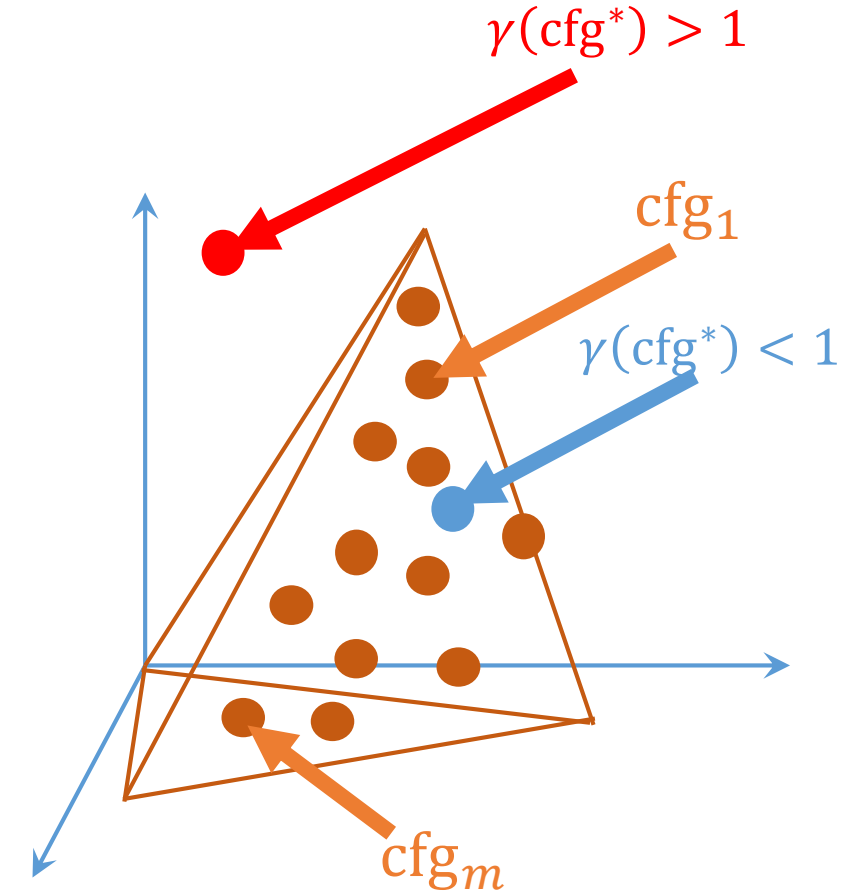
AL with D-optimality criterion: extrapolation grade

Assume we have any **atomistic simulation**: MD, relaxation, etc. In order to decide whether to **consider** a **given configuration** cfg^* as a **candidate** for **inclusion** in the **training set** we introduce the so-called **extrapolation grade**:

$$\gamma(\text{cfg}^*) = \max_{1 \leq j \leq m} (|c_j|), \text{ where}$$

$$\mathbf{c} = \left(\frac{\partial E}{\partial \theta_1}(\boldsymbol{\theta}, \text{cfg}^*) \dots \frac{\partial E}{\partial \theta_m}(\boldsymbol{\theta}, \text{cfg}^*) \right) A^{-1}.$$

This grade defines the **maximal factor** by which the determinant $\det(A)$ can **increase** if cfg^* is **added** to the **active set**, i.e., if $\gamma(\text{cfg}^*) > 1$ then $\det(A)$ could be increased. We say that the potential **extrapolates** if $\gamma(\text{cfg}^*) > 1$, and **interpolates** otherwise.

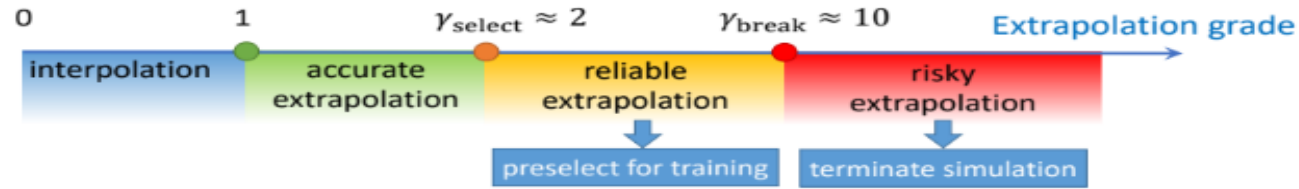


Blue configuration is geometrically close to the ones in the **active set**, the potential **interpolates** on it.

Red configuration is geometrically far from the ones in the **active set**, the potential **extrapolates** on it.

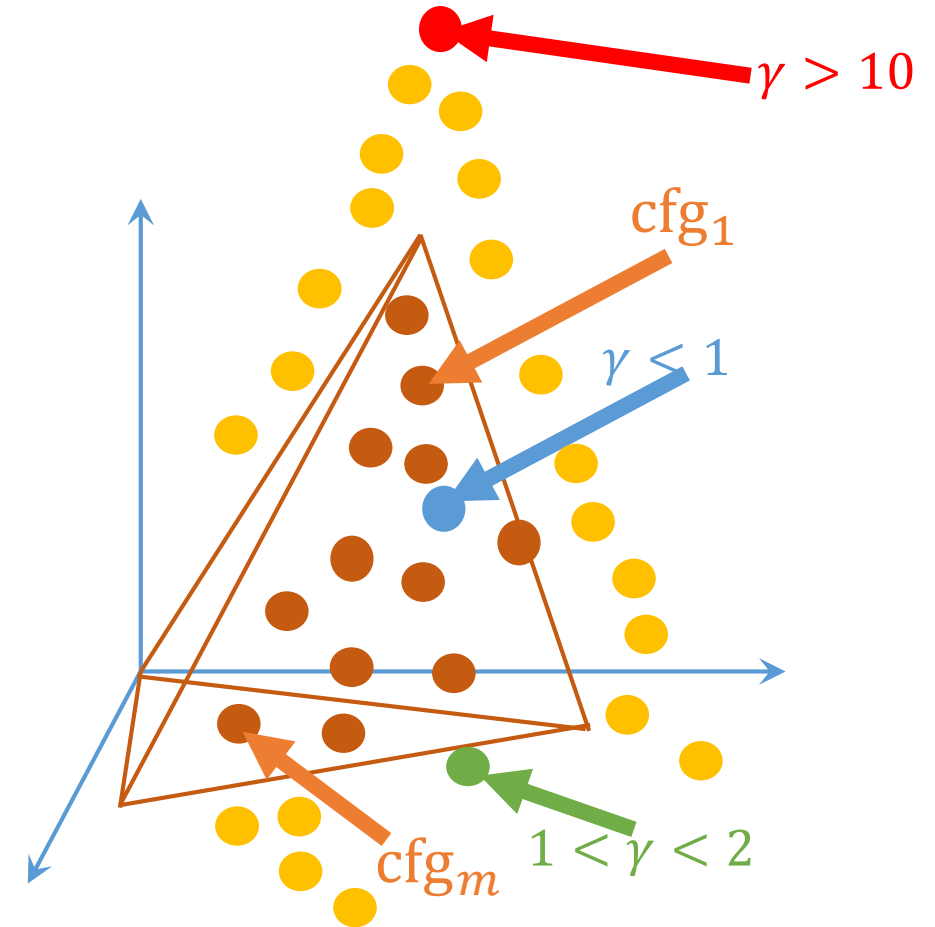
AL with D-optimality criterion: two-threshold scheme

For creating the so-called **preselected set** we choose **two thresholds**: $\gamma_{\text{select}} \approx 2$ and $\gamma_{\text{break}} \approx 10$ [Mach. Learn.: Sci. Technol. 2 (2021) 025002].



If $\gamma_{\text{select}} \leq \gamma(\text{cfg}^*) \leq \gamma_{\text{break}}$ then we **add** the configuration to the **preselected set** and **continue** an atomistic simulation. If $\gamma(\text{cfg}^*) > \gamma_{\text{break}}$ then we **break** an atomistic simulation and **form a matrix** $P \times m$, where P is the number of preselected configurations:

$$B = \begin{bmatrix} \frac{\partial E}{\partial \theta_1}(\boldsymbol{\theta}, \text{cfg}_{m+1}) & \cdots & \frac{\partial E}{\partial \theta_m}(\boldsymbol{\theta}, \text{cfg}_{m+1}) \\ \vdots & \ddots & \vdots \\ \frac{\partial E}{\partial \theta_1}(\boldsymbol{\theta}, \text{cfg}_{m+P+1}) & \cdots & \frac{\partial E}{\partial \theta_m}(\boldsymbol{\theta}, \text{cfg}_{m+P+1}) \end{bmatrix}$$



Yellow and **red** configurations are with **reliable** and **risky** extrapolation, we add them to the **preselected set**.

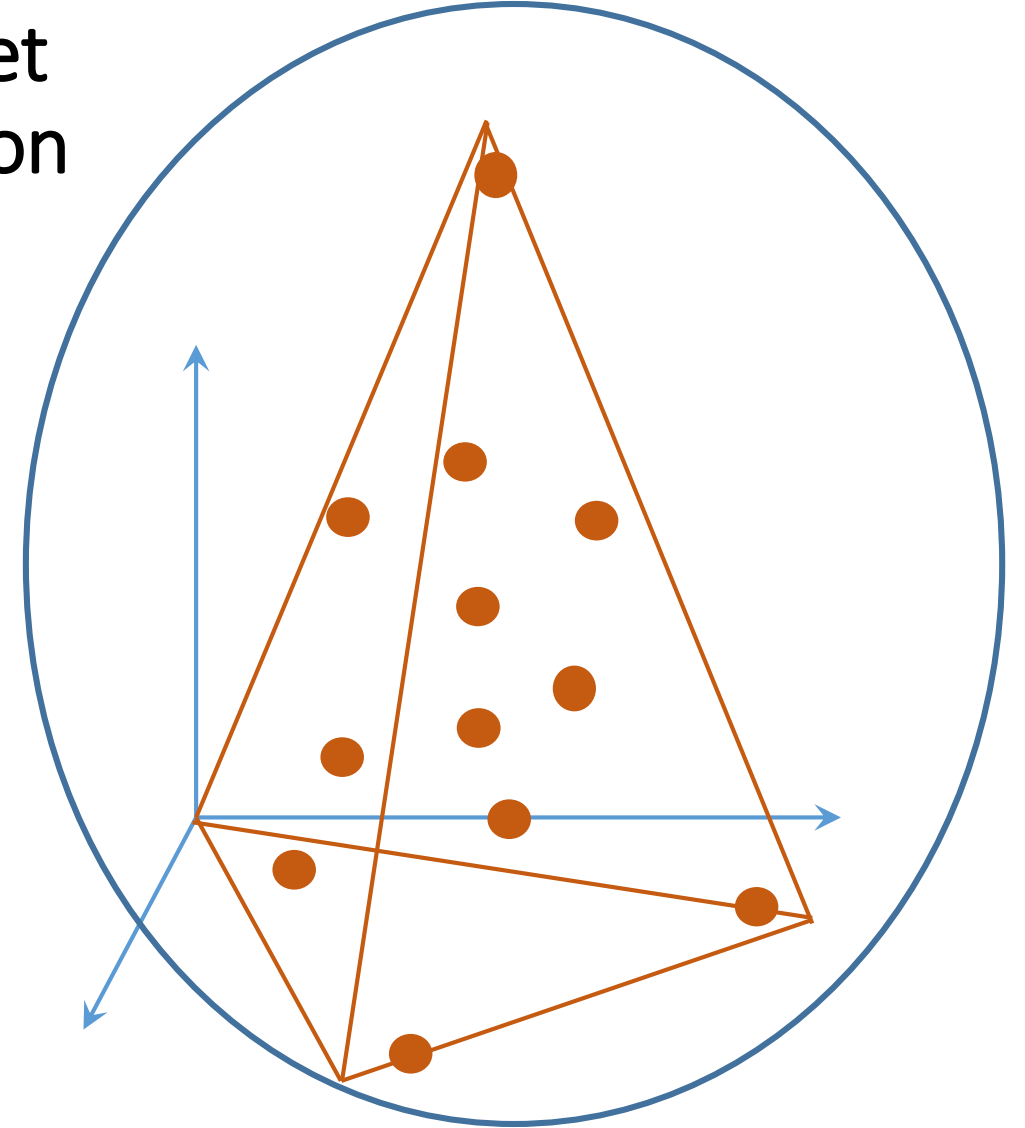
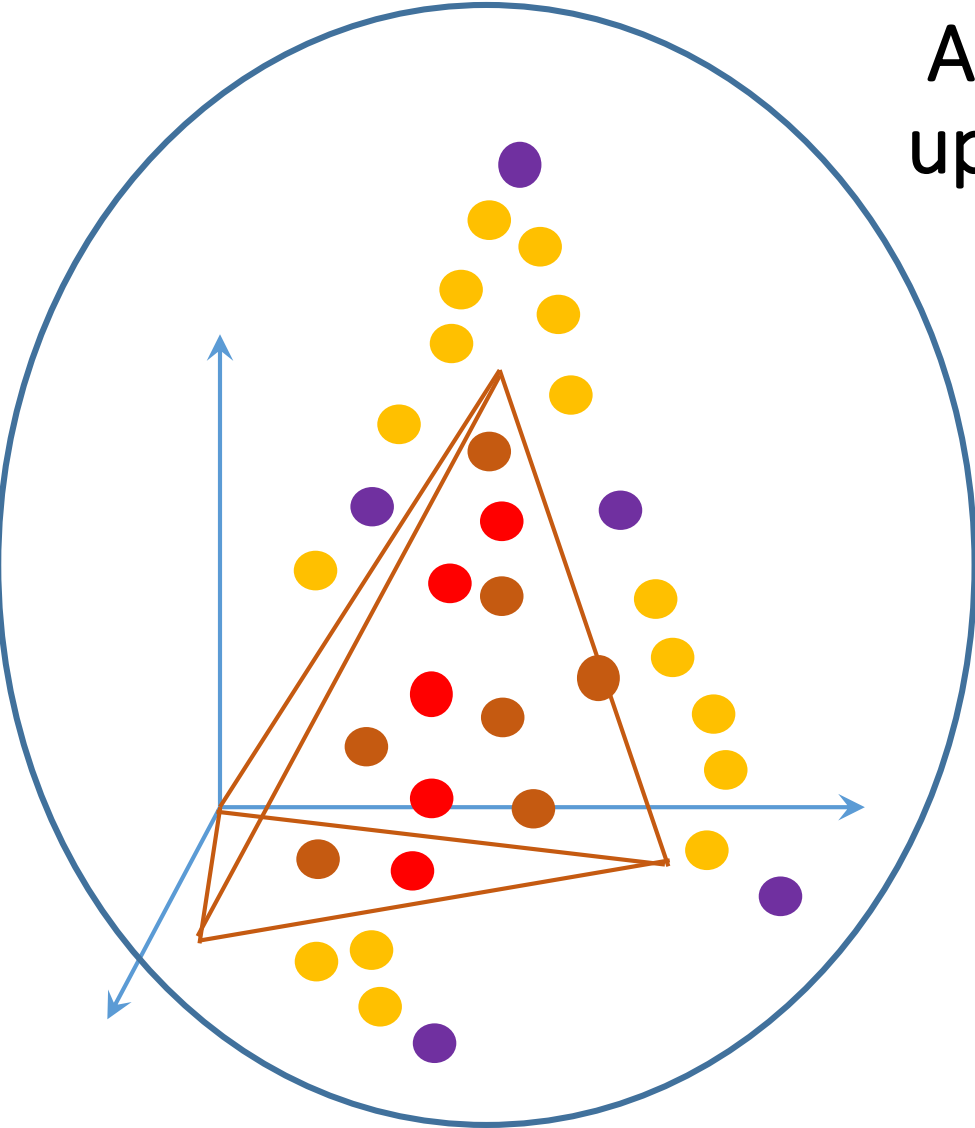
AL: update the active set and the training set

With the **MaxVol algorithm** we **select** the **configurations** that **maximize** the **volume** of the matrix A (or, $\det(A)$). Thus, we **select** $S \leq m$ configurations from the **preselected set**, or, in other words, we **find a submatrix** of **maximum volume** from a **matrix**:

$$\tilde{B} = \begin{bmatrix} \frac{\partial E}{\partial \theta_1}(\boldsymbol{\theta}, \text{cfg}_1) & \cdots & \frac{\partial E}{\partial \theta_m}(\boldsymbol{\theta}, \text{cfg}_1) \\ \vdots & \ddots & \vdots \\ \frac{\partial E}{\partial \theta_1}(\boldsymbol{\theta}, \text{cfg}_{P+m+1}) & \cdots & \frac{\partial E}{\partial \theta_m}(\boldsymbol{\theta}, \text{cfg}_{P+m+1}) \end{bmatrix}$$

where the first m lines are from the matrix A . Next, we use **DFT calculations** for obtaining **energies, forces, and stresses of these configurations**, **append** them to the **training set**. We also **update** the **active set** by substituting S lines in the matrix A . Finally, **we re-train** the potential on the **updated training set**.

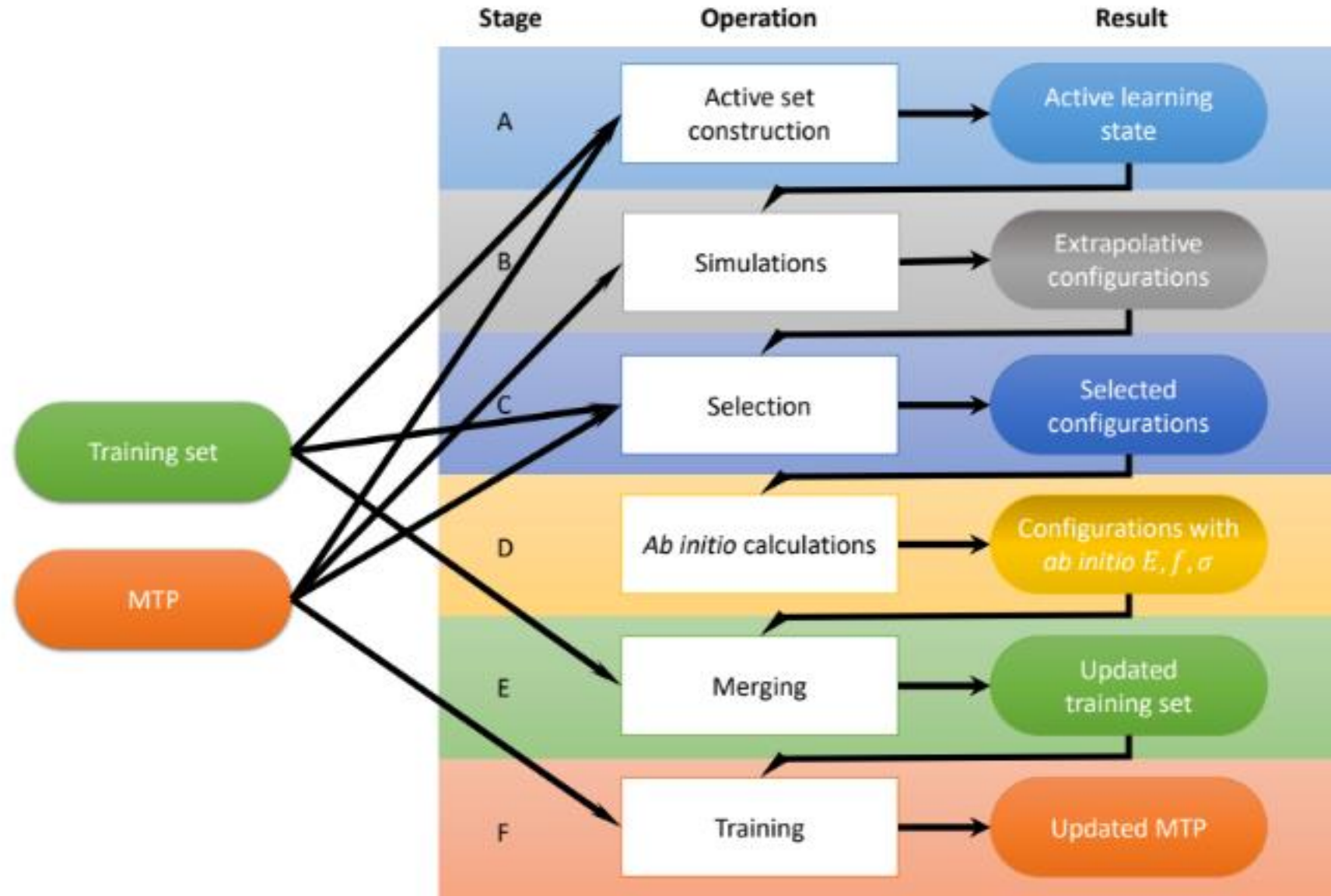
AL: the active set update illustration



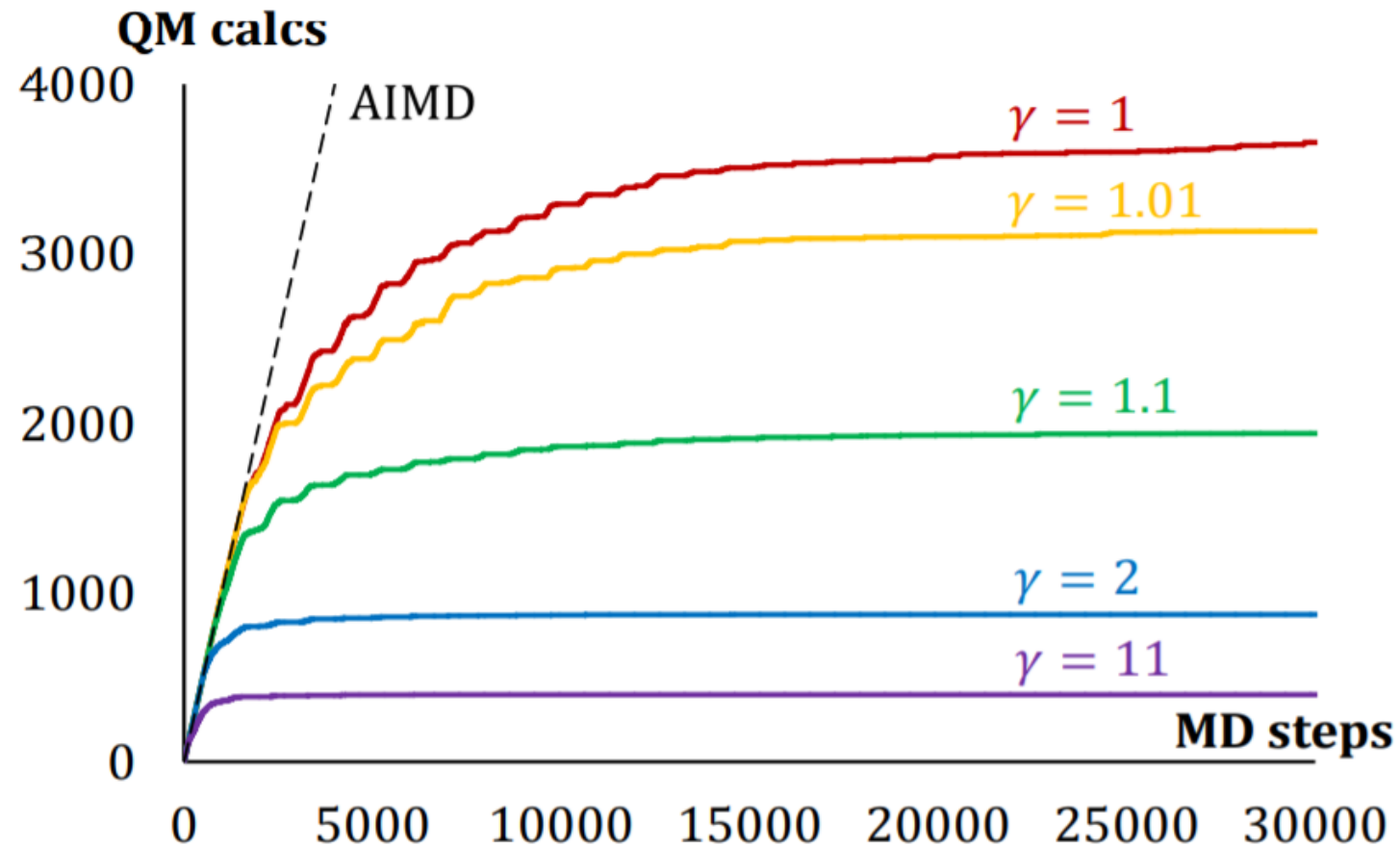
We **select** only the **purple** configurations for updating the training and active sets as they maximize the volume of the pyramid. We **remove** the **red** configurations from the active set and add the **purple** ones instead of them.

The volume of the pyramid (active set) was increased (maximized), the current active set covers a greater part of configuration space than the previous one.

Scheme of active learning iterations

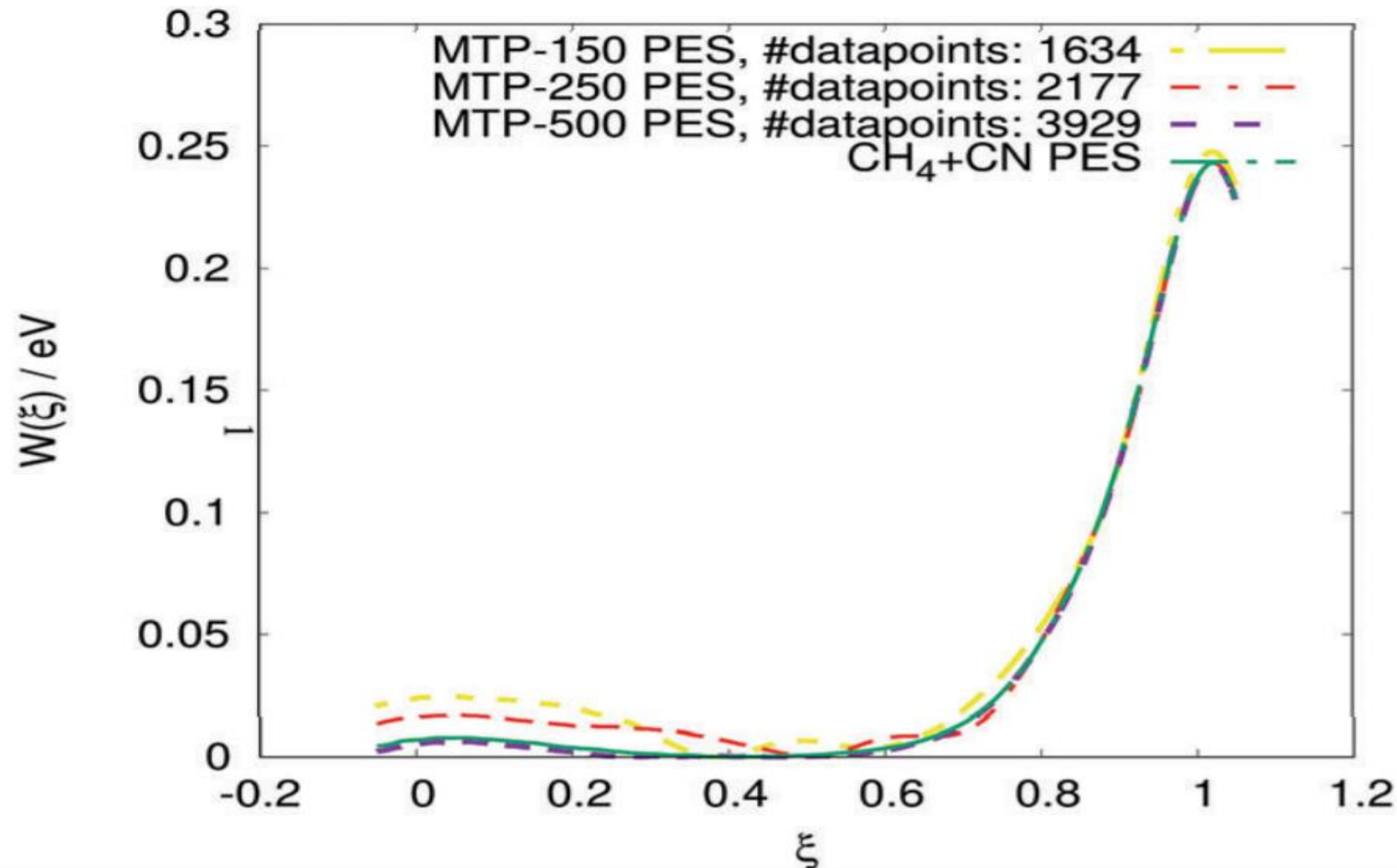


Example: MD in NVT-ensemble for bcc-Li at T=300 K



Number of DFT (QM) calculations decreases while increasing γ_{select} [Computational Materials Science 140 (2017) 171–180].

Example: MD with Andersen thermostat for the chemical reaction $\text{CH}_4 + \text{CN} \rightarrow \text{CH}_3 + \text{HCN}$ at $T=300$ K



Number of configurations in the training set and the predictive power (or, accuracy) of MTP increases while increasing the number of MTP parameters [Phys. Chem. Chem. Phys., 2018, 20, 29503-29512].

Thank you for your attention!

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